



An Introduction to CloudBees

This article introduces the CloudBees PaaS (Platform as a Service) offering.

According to Gartner, the application infrastructure is the mid-tier platform technology layer of the software stack (a.k.a. middleware) where architecture standards, best practices, prevailing protocols and programming models for business applications are defined. Platform as a Service (PaaS) enables organisations to use the application infrastructure as a software service to create, run and integrate cloud-based business applications.

The PaaS resides within the space between Software as a Service (SaaS) and Infrastructure as a Service (IaaS). The latter delivers basic network, storage and compute/processing capabilities as standardised, scalable service offerings. SaaS delivers software capabilities as online Web applications and Web services.

PaaS offerings often support DevOp practices, which include self-service, automated provisioning, continuous integration and continuous delivery. With PaaS, the team can see whether an application is working, broken or staged, and so on—across the entire application lifecycle. Apart from that, PaaS provides:

- On-demand self-service (service catalogues, incremental DevTest)
- Rapid elasticity (provisioning, de-provisioning, flexibility and scalability)
- Resource pooling (platform environments commonly pool memory, code libraries, database connections; multi-tenancy, resource utilisation)
- Measured services (monitoring, metering, billing, etc). Usage is monitored, and the system generates bills based on the charging model.

In the next three to five years, PaaS providers will be capable of providing comprehensive offerings that are not dedicated to a single language.

The Java PaaS

PaaS for Java has come a long way in the past 24 months. PaaS product offerings are rapidly evolving, which is great news for the Java community – there are now low-cost, scalable and hassle-free hosted solutions for Java environments.

The Java platform is well suited for PaaS since the JVM, the application server and deployment archives

provide natural isolation for Java applications, allowing multiple developers to deploy applications in the same infrastructure. Google App Engine was the lone PaaS provider for Java developers. Fortunately, the scenario is changing; it makes sense, since Java developers represent one of the biggest developer communities in the world. In the past two years or so, several commercial providers have entered the Java PaaS space:

- CloudBees
- Amazon Elastic Beanstalk
- CloudSwing
- Cloud Foundry
- Google App Engine
- Heroku
- Red Hat OpenShift

An introduction to CloudBees

CloudBees is run by JBoss and Sun veterans. Its weight is growing in the Java PaaS space. CloudBees brings several unique features into the Java PaaS panorama, in particular, continuous integration—a complete development, testing and deployment management in the cloud.

The CloudBees ecosystem

For JEE developers, I consider that CloudBees and OpenShift offer the ‘best of the breed’ services so far. And between the two, CloudBees is the winner in this highly competitive landscape.

While all other Java PaaS vendors focus on providing a hosted runtime environment for Java applications, CloudBees takes the platform concept further to support the complete development, testing and deployment lifecycle of Java applications.

Features

Supported technology platforms and stacks

One of the most important attributes of CloudBees is the technology platform and stack it supports. CloudBees supports Tomcat, Java SE, Java EE, standard Java libraries, MySQL, commercial relational databases and Big Data. It allows file system and thread access. No special framework is